LAB EXPERIMENT RECORD

Set 24: Product Inventory Management

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. Bar Chart

Aim

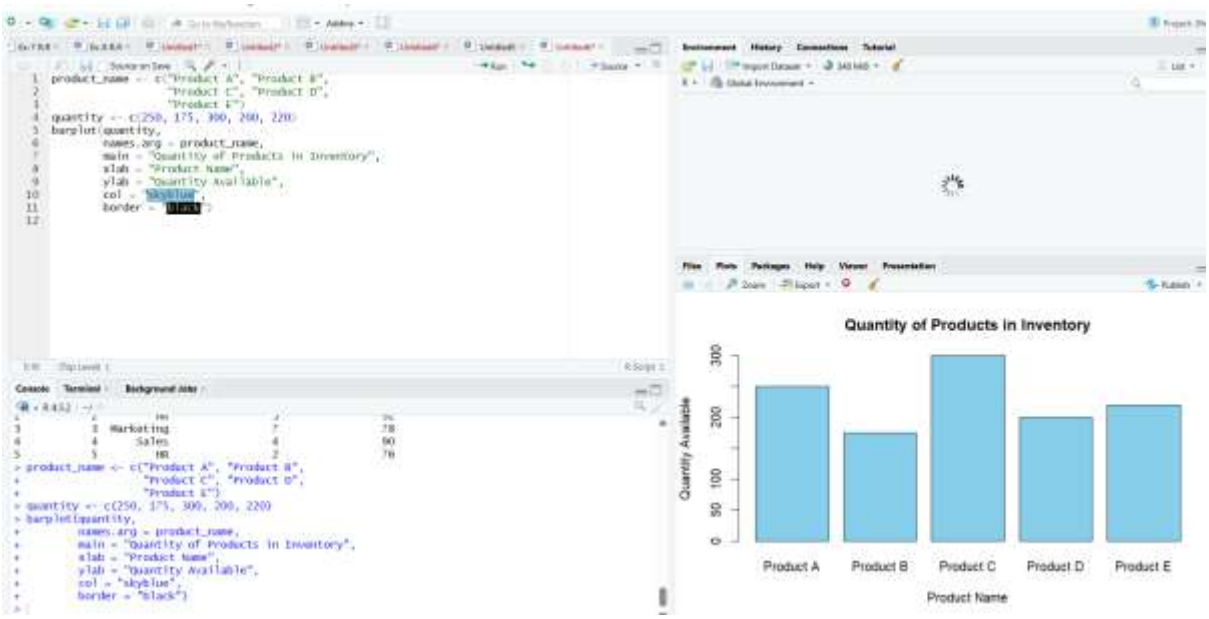
To create a bar chart in R to visualize the quantity of each product in the inventory and label the axes and title the chart.

R Code

R

```
product_name <- c("Product A", "Product B",
                  "Product C", "Product D",
                  "Product E")
quantity <- c(250, 175, 300, 200, 220)
barplot(quantity,
        names.arg = product_name,
        main = "Quantity of Products in Inventory",
        xlab = "Product Name",
        ylab = "Quantity Available",
        col = "skyblue",
        border = "black")
```

Output



A bar chart showing the available quantities of all five products.

Observation

Product C has the highest quantity of 300 units, while Product B has the lowest quantity of 175 units.

2. Stacked Bar Chart

Aim

To generate a stacked bar chart showing the quantity of each product within different product categories.

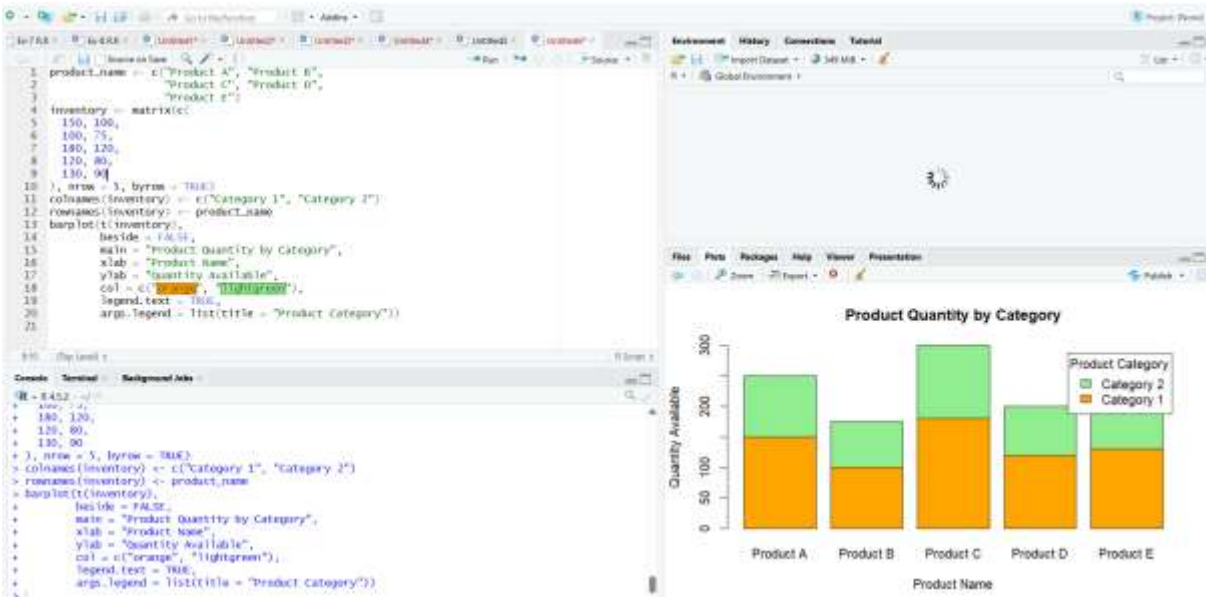
R Code

R

```
product_name <- c("Product A", "Product B",
                  "Product C", "Product D",
                  "Product E")
inventory <- matrix(c(
  150, 100,
```

```
  100, 75,
  180, 120,
  120, 80,
  130, 90
), nrow = 5, byrow = TRUE)
colnames(inventory) <- c("Category 1", "Category 2")
rownames(inventory) <- product_name
barplot(t(inventory),
        beside = FALSE,
        main = "Product Quantity by Category",
        xlab = "Product Name",
        ylab = "Quantity Available",
        col = c("orange", "lightgreen"),
        legend.text = TRUE,
        args.legend = list(title = "Product Category"))
```

Output



A stacked bar chart displaying the quantity of each product divided into two categories.

Observation

Product C has the highest total quantity of 300 units, followed by Product A with 250 units. The category-wise quantities are assumed for demonstration because the original dataset does not provide product categories.

3. Tableau Dashboard

Aim

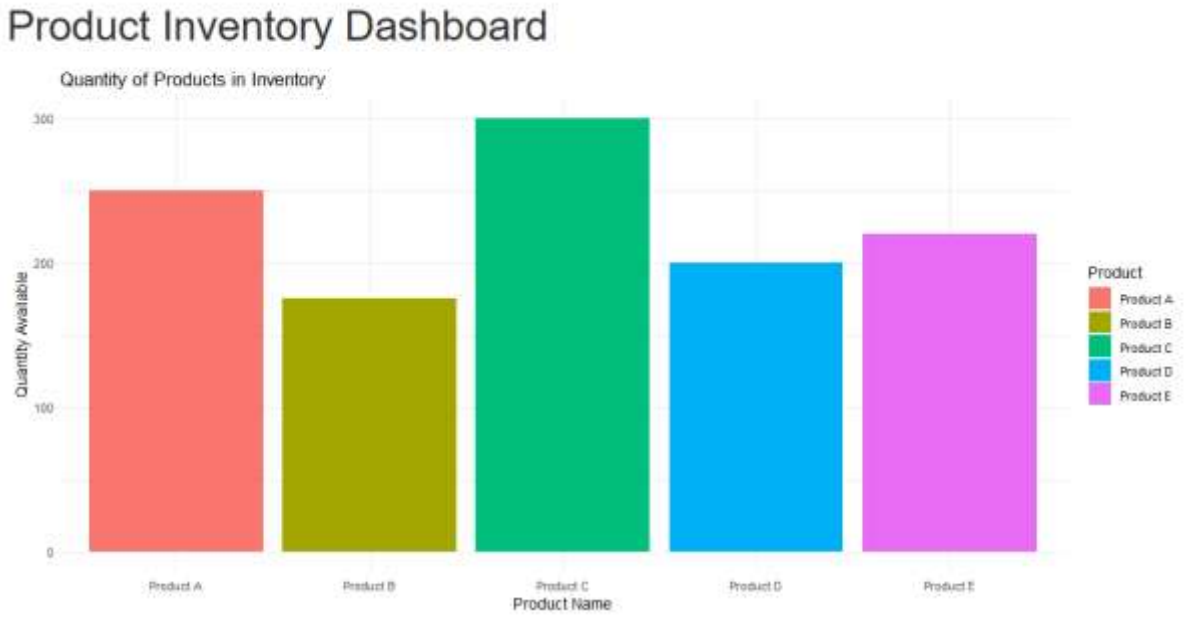
To develop an interactive dashboard combining a bar chart and a stacked bar chart for exploring inventory data.

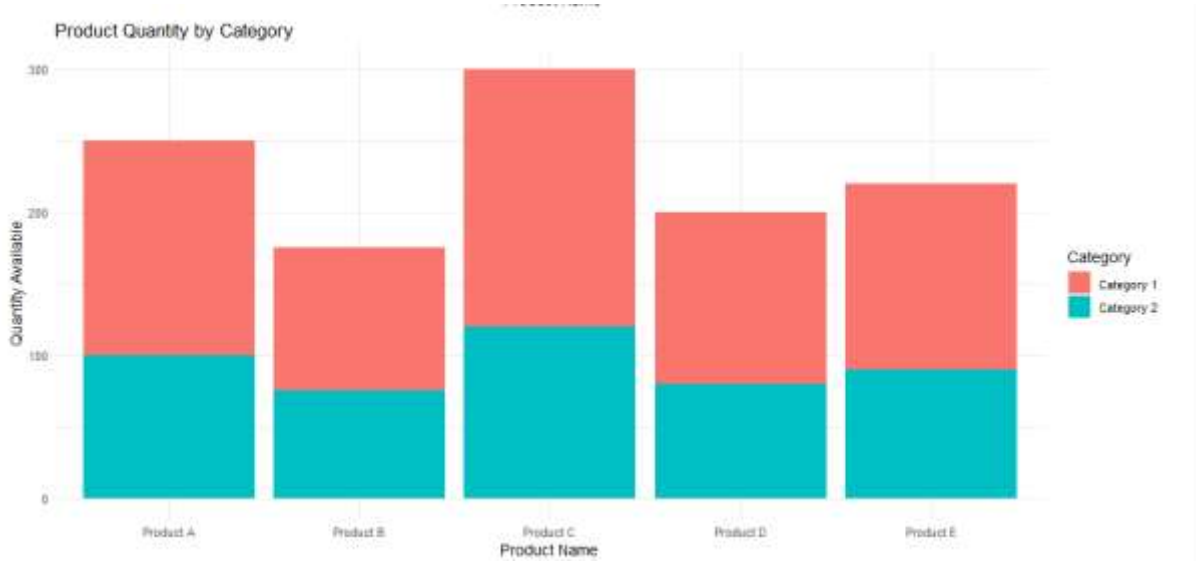
Tableau Procedure

1. Open Tableau and import the Product Inventory dataset.
2. Create a bar chart using Product Name and Quantity Available.
3. Add a suitable chart title and axis labels.
4. Create a separate category dataset containing the assumed quantities for Category 1 and Category 2.
5. Create a stacked bar chart using Product Name, Quantity Available, and Product Category.

6. Assign different colors to the product categories.
7. Select **Dashboard → New Dashboard**.
8. Drag the inventory bar chart into the dashboard.
9. Drag the stacked bar chart into the dashboard.
10. Arrange both charts neatly.
11. Add suitable titles, labels, and filters for interactive exploration.
12. Save the dashboard.

Output





An interactive dashboard displaying product quantities and category-wise inventory quantities.

Observation

The dashboard helps users compare the total quantities of products and examine their distribution across different categories.

4. Scatter Plot: Product Price vs Quantity Available

Aim

To create a scatter plot in R to explore the relationship between product price and quantity available.

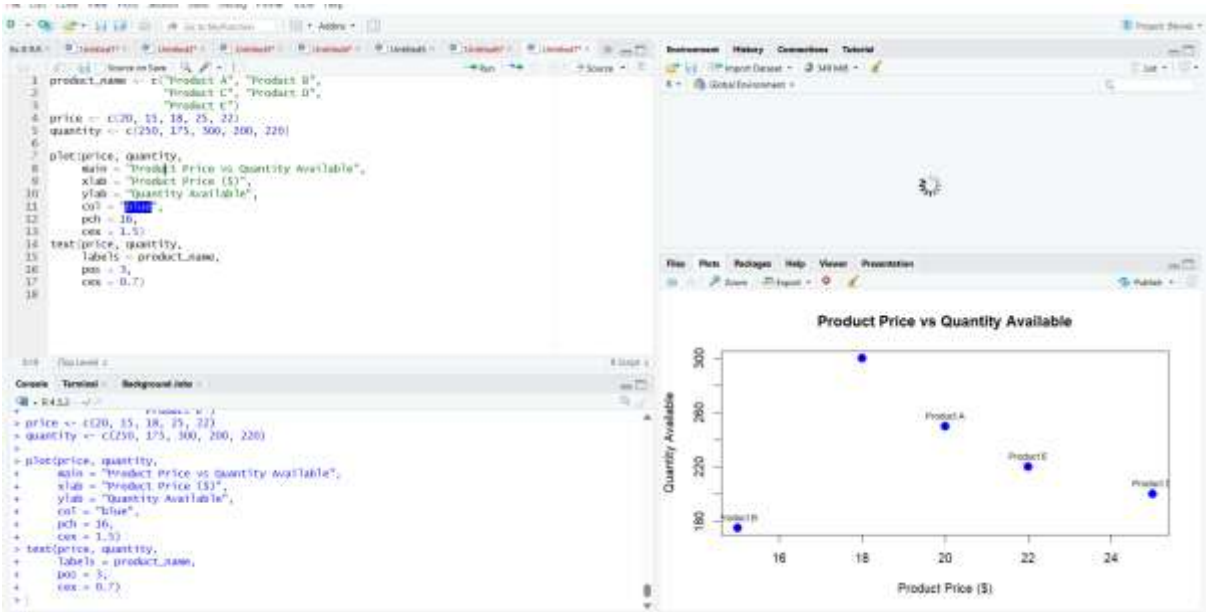
R Code

R

```
product_name <- c("Product A", "Product B",
                  "Product C", "Product D",
                  "Product E")
price <- c(20, 15, 18, 25, 22)
quantity <- c(250, 175, 300, 200, 220)
```

```
plot(price, quantity,
     main = "Product Price vs Quantity Available",
     xlab = "Product Price ($)",
     ylab = "Quantity Available",
     col = "blue",
     pch = 16,
     cex = 1.5)
text(price, quantity,
     labels = product_name,
     pos = 3,
     cex = 0.7)
```

Output



A scatter plot showing the relationship between product prices and available quantities.

Observation

Product C has the highest available quantity of 300 units at a price of $18. Product B has the lowest quantity of 175 units at a price of $15. The data does not show a clear consistent relationship between price and quantity. The prices are assumed for demonstration because they were not provided in the original dataset.

Result

The product inventory dataset was analyzed using a bar chart, stacked bar chart, and scatter plot in R. A Tableau dashboard was also outlined to combine the inventory visualizations for interactive exploration.